Dr. Princy Ranaivomanana

Ph.D. in Astronomy & Astrophysics

Time-Domain Photometry · Space- & Ground-Based Observatories · Machine Learning for Variability Analysis

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Research Profile

Observational and data-driven astronomer specialising in the variability of compact evolved stars. My PhD focused on photometric characterisation of hot sub-luminous stars using Gaia observations, combined with multi-colour photometry from BlackGEM. I am experienced in frequency analysis of irregular and sparsely sampled time-series data, machine learning, and the development of open-source Python pipelines for large photometric surveys.

Research & Project Experience

2026 – Present **Visiting researcher**, University of Antananarivo, *Madagascar*

10/2021 – 09/2025 **PhD Research Project**, Radboud University · KU Leuven, *The Netherlands / Belgium*

- Identified 193 new hot subdwarf variables and filtered contaminant sources from a catalogue of ∼60,000 objects using unsupervised machine learning (UMAP, t-SNE) on Gaia DR3 time-series photometry.
- Designed and implemented signal processing and frequency analysis pipelines to detect periodic signals in sparsely sampled, heteroscedastic time-series data from Gaia, TESS, and BlackGEM.
- Contributed to multi-colour photometry of the DOV star PG 1159–035 with the BlackGEM array, demonstrating hands-on experience with space- and ground-based photometric instrument commissioning and data reduction.

2020 – 2021 **Master Research Project**, University of Manchester, *United Kingdom*

- Developed semi-supervised GAN models for pulsar data synthesis and classification, with relevance to large radio surveys (SKA), demonstrating experience with automated classification pipelines for high-volume survey data.

Feb 2026 **Modelling Camp**, Radboud University, *The Netherlands*

- Collaborated in a multidisciplinary team to develop machine learning solutions for the Nutreco company.

Education

10/2021 – 09/2025 **Dual PhD in Astronomy & Astrophysics**, Radboud University – KU Leuven, *The Netherlands / Belgium*

Thesis: *Hot subluminous stars: Time-series analysis and pulsation mode identification*
Supervisors: Prof. Paul Groot, Prof. Conny Aerts, Dr. Cole Johnston

2020 – 2021 **Master of Science in Astronomy**, University of Manchester, *United Kingdom*

Thesis: *"Building a deep generative model for surveyed pulsar classes"*

2015 – 2017 **Master of Science in Astronomy**, University of Antananarivo, *Madagascar*

Thesis: *"Young massive star clusters hosted by luminous infrared galaxies, Arp 299"*

2012 – 2015 **B.Sc. in Physics**, University of Antananarivo, *Madagascar*

Teaching, Supervision, Mentoring

05/2026 – Present **Career mentor**, CorrelAid Mentoring program, *Germany*

Advised a Master's student on industry and academic career pathways

2023 – 2024 **Teaching Assistant**, Radboud University, *The Netherlands*

Tutorials: Data Analysis (2021, 2024) and Astronomical Instrumentation (2023–2024)

2022 **Bachelor Project Supervisor**, KU Leuven, *Belgium*

Supervised two bachelor research projects (two students each)

2023 **Teaching Course**, Radboud University (certificate)

2022 **Teaching Assistant Training**, KU Leuven

Peer-Reviewed Publications

8 refereed papers (5 first-author) · h-index & citations: verify at NASA/ADS · ORCID: 0000-0001-8470-0220

[1] **P. Ranaivomanana** and M. Uzundag, "Exploring late stellar evolution in the era of large surveys: Machine learning prospects for hot subdwarfs and white dwarfs," *In production, submitted to MDPI Universe*

[2] **P. Ranaivomanana**, C. Johnston, M. Uzundag, P.J. Groot, C. Aerts, T. Kupfer, "BlackGEM observations of compact pulsating stars: mode identification for the DOV PG 1159−035 using multi-colour photometry," *A&A, 708, A106 (2026)*

[3] **P. Ranaivomanana**, C. Johnston, G. Iorio, P.J. Groot, M. Uzundag, T. Kupfer, C. Aerts, "Unsupervised learning for variability detection with Gaia DR3 photometry: the main sequence–white dwarf valley," *A&A, 704, A70 (2025)*

[4] **P. Ranaivomanana**, M. Uzundag, C. Johnston, P.J. Groot, T. Kupfer, C. Aerts, "Variability in hot sub-luminous stars and binaries: machine-learning analysis of Gaia DR3 multi-epoch photometry," *A&A, 693,*

A268 (2025)

[5] **P. Ranaivomanana**, C. Johnston, P.J. Groot, C. Aerts, R. Lees, L. IJspeert, S. Bloemen, M. Klein-Wolt, P. Woudt, E. Körding, R. Le Poole, D. Pieterse, "Identifying and characterising the population of hot sub-luminous stars with multi-colour MeerLICHT data," *A&A, 672, A69 (2023)*

Talks & Conference Contributions

Invited talks

2025 — **Hamburg Observatory Colloquium**, Hamburg, Germany

Unsupervised machine learning for variability analysis in time-domain astronomy

Contributed talks

2025 — **12th International Meeting on Hot Subdwarfs & Related Objects**, North Carolina, USA

Unravelling Hot Subdwarfs and Binaries Variability with Gaia DR3 and Machine Learning

2024 — **KOPAL 2024 (flash talk)**, Litomyšl, Czech Republic

2023 — **MW-Gaia WG2 Conference**, Sofia, Bulgaria

Classification of variable stars observed in multiple filters with MeerLICHT and BlackGEM

Posters

2024 — **TASC/KASC 15**, Porto, Portugal

2023 — **AMCVn5 workshop & sdOB11 conference**, Armagh, UK

2022 — **TASC/KASC 13 & sdOB10 conference**, Leuven / Liège, Belgium

Schools

2023, 2025 — **Oxford Machine Learning School (OxML)**, Oxford, UK

Technical Skills

Survey Photometry	*Gaia*, TESS, ZTF, BlackGEM, MeerLICHT: light-curve extraction, detrending, aperture and PSF photometry
Signal Processing	Lomb-Scargle periodogram, Fourier analysis, periodicity search for heteroscedastic time-series data
Programming	Python (Astropy, NumPy, SciPy, Pandas, emcee, scikit-learn, TensorFlow, Keras, PyTorch), SQL, Bash scripting, LaTeX
Machine Learning	Unsupervised clustering (UMAP, t-SNE), semi-supervised GANs, classification and anomaly detection on tabular/time-series/image data
Data Pipelines	Automated quality-control workflows, job scheduling (SLURM), Git/GitHub, Docker

Grants, Awards & Prizes

2023 / 2024 / 2025 — **Leids Kerkhoven-Bosscha Fonds (LKBF) Conference Grant**, Leiden Observatory

Three consecutive grants to attend international conferences

2020 — **Newton Fund / DARA Big Data MPhil Bursary**, UK Space Agency / UK Research & Innovation

Full bursary for MPhil studies at the University of Manchester

2019 — **DARA Travel Grant**, African Radio Interferometry Winter School, South Africa

2018 — **SKA / SARAO Grant**, 2nd Big Data Africa School, South Africa

2017 — **IAU Travel Grant**, 39th International School for Young Astronomers (ISYA), Ethiopia

Relevant Certifications

- Machine Learning in Production — *DeepLearning.AI / Coursera*
- Advanced Machine Learning and Signal Processing — *Coursera*
- Generative AI: Foundation Models and Platforms — *Coursera*
- Exploratory Data Analysis for Machine Learning — *Coursera*
- Data Analysis with Python — *Coursera*
- Data Fundamentals — *IBM SkillsBuild*
- Teaching Training Course — *Radboud University*

Outreach & Service

2023 — **BlackGEM Telescope Commissioning**, ESO La Silla Observatory, Chile

Two-week on-site contribution to commissioning the BlackGEM array — hands-on experience with ground-based telescope operations and data verification

2025 — **High School Mentoring**, The Netherlands

Guided high school students through time-series analysis projects

2018 – 2019 — **Project Leader**, Madagascar Astronomy Magazine (OAD/DARA-funded)

2019 — **National Coordinator**, Malagasy Astronomical Society (MAS) & IAU100 national activities